

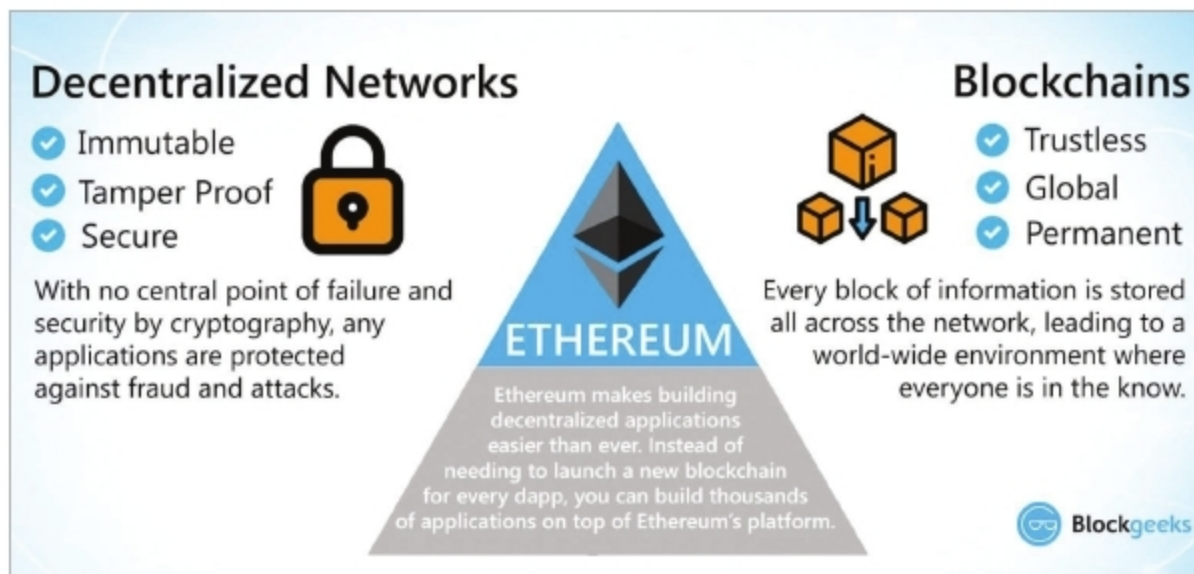


Figure 2: Advantages of Ethereum's decentralised system (Source: BlockGeeks)

the 'smart contract', a creation of this blockchain's founders. Buterin and his co-founders launched a crowdsourcing campaign, under which they sold the participants *Ether* to raise funds to the tune of US\$ 18 million. Armed with the funds and a solid vision, they first released Ethereum back in 2015. This first, barebones implementation was known as the Frontier. The platform saw considerable success and now has a widespread community working on it. One of the main selling points of this technology is that it offers support for generalised scripting and application development, without being limited to only financial applications. By virtue of its uniqueness, expansive reach and adoption rate, Ethereum has become one of the frontrunners of the blockchain revolution — driven by expanding potential applications and distributed consensus.

Ethereum's long-term vision, as stated on its website, is to 'decentralise the Internet' and make it 'the way it was supposed to be'. A smart contract is described by the site as "an application that runs exactly as programmed without downtime, censorship, fraud or third party interference." The platform is built around the idea of smart contracts that allow anyone to write decentralised Web applications as long as they have a basic understanding of distributed consensus. This has opened the doors for various non-technical sections of society to benefit from finding their own uses for this marvellous technology.

The story of Ethereum includes interesting twists and turns. After the first implementation in 2015, it once again began as the decentralised autonomous organisation (DAO) that

raised US\$ 150 million in a crowd sale to fund the project in 2016. It then had to undergo a series of hard forks in order to reverse the robbery of US\$ 50 million in *Ether* by a hacker, and subsequent attacks. Eventually, the team debloated the blockchain and added protection against DDoS attacks. However, this resulted in the creation of differing versions, including Ethereum (on the forked blockchain) and Ethereum Classic (on the original blockchain).

Major attacks

It is important to understand the flaws of cryptocurrencies if we are to truly estimate their potential to work in multi-faceted applications, at scale. One of the first major attacks on Ethereum was the result of a recursive call bug within the code that allowed users of the smart contract to withdraw twice the *Ether* they invested within the contract. It was brought to the developers' notice but they claimed it was a known bug and no *Ether* was at risk. Unfortunately, they were mistaken and the platform was exploited by a hacker who went so far as to write an open letter claiming that the US\$ 50 million he stole was legal and within the terms of the smart contract. He even threatened legal action should the Ethereum team attempt to reverse this transfer. Needless to say, the developers went ahead and created the first hard fork, reversing the loss of this money and allowing the affected investors to withdraw their original investment.

To understand how attackers tried to exploit the Ethereum platform, let's take a closer look at the flaws in the setup. Much like your vehicles, the Ethereum virtual machine runs on *gas*, which is a fraction of an *Ether* and is used to make transactions on the Ethereum blockchain. Initially, some transactions on the platform cost very little *gas*. So little, that it opens the doors for exploitation, as I will explain now.



Figure 3: A history of Ethereum's price (Source: CoinMarketCap)